

Well-Calibrated Posteriors
Under Model Misspecification
via β -Bayes

by

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Abstract

Being a widely used statistical framework, Bayesian inference rests on several assumptions, including that the likelihood is specified correctly. In practice, however, this assumption is often violated. This may cause posteriors computed on independent datasets of the same data generating process to significantly disagree, having substantially no overlap. This can lead to unreliable and unstable inference. In this thesis we introduce the way to quantify the amount of disagreement between posteriors and establish a target amount of overlap – how much the posteriors computed on the independent datasets of the same data generating process should overlap. We show that standard Bayesian inference violates this threshold under model misspecification, and introduce a method which can recover well-calibrated posteriors under the misspecified model. We analyze the proposed methodology theoretically and validate it through several experiments.

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1 Introduction

Decision-making in spheres such as sciences, industry, and public policy, relies on data. To draw reasonable and useful conclusions about the various real world processes, we need a principled way to analyze and examine the data — this is the role of statistical inference. The goals of statistical inference can be diverse — from estimating the parameters of interest, to predicting unseen data points based on historical information.

Bayesian inference is a statistical framework which is widely used in many modern-day applications. Bayesian inference uses the assumed model, referred to as the *likelihood* $p(x_{1:N} \mid \theta)$, together with *prior* beliefs $\pi(\theta)$ about the parameter of interest θ , to incorporate newly observed data $x_{1:N}$ and produce *posterior* beliefs

$$\pi(\theta \mid x_{1:N}) = \frac{p(x_{1:N} \mid \theta)\pi(\theta)}{\int p(x_{1:N} \mid \theta)\pi(\theta)d\theta}. \quad (1)$$

Bayesian inference is very appealing both practically and theoretically. Compared to the frequentist approach, it incorporates the already existing domain knowledge in the form of a prior $\pi(\theta)$, returns a distribution rather than a point estimate, directly representing the uncertainty, and allows one to work with more complex model types (Gelman et al., 2013). However, Bayesian inference relies on several assumptions, including the correct specification of the prior, availability of infinite computing power, and, most relevant for this work, the assumption that the likelihood is *well-specified* (Knoblauch, Jewson, and Damoulas, 2022). Formally, we say that the likelihood $p(x \mid \theta)$ is well-specified if the true data-generating process (DGP) $P_\circ$ belongs to the assumed model family, i.e., there exists a $\theta_\circ \in \Theta$, where Θ is the parameter space, such that

$$P_\circ = P_{\theta_\circ},$$

where $P_{\theta_\circ}$ denotes the distribution with density $p(x \mid \theta_\circ)$. In this case, $\theta_\circ$ is the true parameter value and the likelihood correctly reflects the true DGP.

However, in practice, this assumption is often violated (Bissiri, Holmes, and Walker,

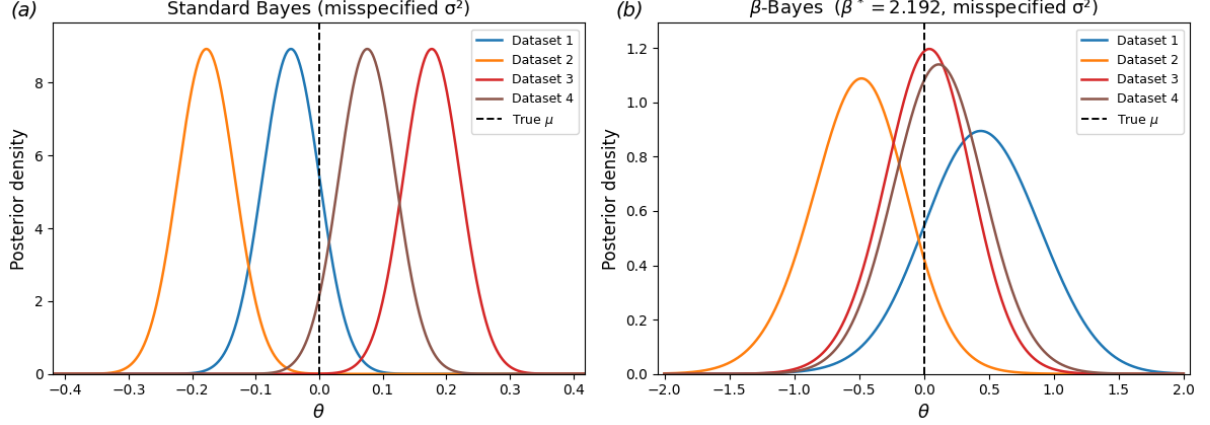


Figure 1: Posterior distributions for four independent datasets of size $N = 500$ drawn from the true DGP $x \sim \mathcal{N}(0, 3.0^2)$, under a misspecified likelihood $x \mid \theta \sim \mathcal{N}(\theta, 1.0^2)$. **(a)** Some pairs of standard posterior distributions have essentially no overlap. **(b)** β -Bayes posteriors at β^* overlap substantially.

2016; Knoblauch, Jewson, and Damoulas, 2022; Huggins and Miller, 2024; Huggins and Miller, 2023). When no such θ_o exists, meaning that

$$\forall \theta \in \Theta, P_o \neq P_\theta,$$

the likelihood is said to be *misspecified*. In this case, under regularity conditions, Bayesian posterior concentrates around the *pseudo-true* parameter

$$\theta_* = \arg \min_{\theta \in \Theta} \text{KL}(P_o \parallel P_\theta), \quad (2)$$

the parameter value closest to the true distribution in Kullback–Leibler (KL) divergence.

Under model misspecification, Bayesian posteriors can be poorly calibrated, meaning that they incorrectly represent the uncertainty about the parameter of interest. This results in lack of posterior reproducibility, across independent data realizations of the same true DGP under the same misspecified model (Huggins and Miller, 2024). The illustration of this problem can be seen on Figure 1 (a).

This problem has important real-world implications. Consider two researchers who independently collect data from the same underlying phenomenon and conduct Bayesian inference using the same model. Intuitively, their posterior distributions should be somehow compatible – if the two posteriors place their mass on disjoint sets of the parameter

space, at least one of the researchers is drawing misleading conclusions. However, some degree of disagreement between the two posteriors is natural and expected, since the two datasets are different realizations of the same DGP. This raises a question: how much disagreement is acceptable? Specifically, how much of the observed disagreement between two posteriors can be explained by the dataset variability, and how much by the distorting effect of model misspecification?

The motivation for our work is given in Huggins and Miller (2024), where it is shown that under misspecification Bayesian posteriors computed on independent data realizations of the same DGP might have essentially no overlap. The authors introduce a criterion which establishes the lower bound on the probability with which the posterior credible sets should overlap, based on the frequentist coverage: if two credible sets each have coverage $1 - \alpha$, then the probability that they intersect must be at least $(1 - \alpha)^2$. They show that under misspecification, standard Bayes can violate this bound, while the methodology they propose, *BayesBag*, typically satisfies it.

In this study we develop a criterion for the expected overlap between the two posteriors $\pi(\theta \mid x_{1:N})$ and $\pi(\theta \mid y_{1:N})$, where $x_{1:N}$ and $y_{1:N}$ are independently drawn from the true DGP $P_\circ$. In other words, we quantify how much $\pi(\theta \mid x_{1:N})$ and $\pi(\theta \mid y_{1:N})$ should overlap, show that under model misspecification standard Bayes frequently violates this threshold, and introduce a method to obtain posteriors that satisfy it.

The solution we propose is based on the framework developed by Bissiri, Holmes, and Walker (2016) and further explored by Jewson, Smith, and Holmes (2024) and Knoblauch, Jewson, and Damoulas (2022), which is called *Generalized Bayesian Inference* (GBI). GBI allows to update prior beliefs $\pi(\theta)$ after observing data $x_{1:N}$ even when the likelihood does not exactly capture the true data-generating distribution $P_\circ$. Specifically, rather than the standard posterior update, the generalized posterior under GBI is defined as

$$\pi^{(\ell)}(\theta \mid x_{1:N}) = \frac{\pi(\theta) \exp\left(-w \sum_{i=1}^N \ell(\theta, x_i)\right)}{\int \pi(\theta) \exp\left(-w \sum_{i=1}^N \ell(\theta, x_i)\right) d\theta}, \quad (3)$$

where $\ell(\theta, x)$ is a specified loss function and $w > 0$ is a scaling parameter. This update

targets the parameter

$$\theta_*^{(\ell)} = \arg \min_{\theta \in \Theta} \mathbb{E}_{P_\circ}[\ell(\theta, x)], \quad (4)$$

without requiring that θ indexes a correctly specified model for $P_\circ$. Clearly, when $w = 1$ and negative log-likelihood is chosen as the loss function ℓ , (3) recovers (1).

As an alternative to negative log-likelihood, we use (3) with the β -divergence loss, introduced by Basu et al. (1998), and defined pointwise as

$$\ell^{(\beta)}(\theta, x_i) = -\frac{1}{\beta - 1} p(x_i | \theta)^{\beta-1} + \frac{1}{\beta} \int p(z | \theta)^\beta dz. \quad (5)$$

We refer to GBI with β -divergence loss as β -Bayes. When $\beta \rightarrow 1$, (5) recovers negative log-likelihood.

The β -divergence loss is one of the most widely used robust loss functions, being an alternative to the negative log-likelihood. It reduces the influence of observations that are unlikely under the assumed model on the posterior update, downweighting the outliers. Specifically, if some observation x_i is unlikely, meaning that $p(x_i | \theta)$ has near-zero magnitude, raising it to the power $\beta - 1 > 0$ will make its contribution to the posterior even smaller. In particular, the contribution of x_i is proportional to $p(x_i | \theta)^{\beta-1}$, which tends to zero as $\beta \rightarrow \infty$. Therefore, increasing the value of β downweights the influence of outliers and increases robustness. However, there is a trade-off: larger values of β downweight nearly all observations, so the posterior learns less from the data and is dominated by the prior. Additionally, unlike many other robust divergences such as the Hellinger distance or total variation distance, β -divergence loss does not require a non-parametric density estimate of the true DGP (Jewson, Smith, and Holmes, 2024), making it applicable in high-dimensional settings where such estimation is unreliable.

In this study we theoretically show that asymptotically the criterion for overlap is often violated under standard Bayes with model misspecification, and that under β -Bayes, a set of sufficient regularity conditions on the likelihood together with a mild additional condition on the functional of interest, guarantee the existence of a value β^* for which the

overlap criterion is satisfied. We empirically verify the theoretical results on synthetic and real data, propose a procedure for selecting the optimal β^* , and demonstrate that using β -Bayes with β^* has a positive impact on the stability of both parameter inference and prediction, while leaving predictive accuracy roughly unchanged. In Figure 1 (b) we can see that posteriors overlap more substantially under β -Bayes in case of a simple Gaussian location model with a misspecified variance. The methodology we propose allows one to resolve the problem of poorly calibrated posteriors under model misspecification, making the inference more stable to the choice of data realizations of the true DGP.

2 Background and Related Work

Several works target robust inference under model misspecification, some of them pointing out the particular benefits of using β -Bayes, for its increased robustness to outliers as compared to standard Bayes. Jewson, Smith, and Holmes (2024) show that under β -Bayes, stability of posterior predictives holds under feasibly verifiable conditions: stability to the choice of likelihood requires that the two candidate likelihood models are close in total variation, and stability to the DGP requires that the two data-generating processes are close in total variation. Under standard Bayes, the conditions sufficient for stability require closeness in log-probability — of the likelihoods in the first case, of the DGPs in the second — which is almost impossible to verify on practice, as it requires correct specification of tail probabilities. A special case of their framework shows that when the posterior predictives are computed on two different data realizations of the same DGP, the β -divergence between them is bounded from above by the quantity that depends on the amount of model misspecification. Jewson, Smith, and Holmes (2024) target feasibility of conditions under which the amount of overlap between posterior predictives is bounded. We, instead, focus on the amount of overlap between posteriors. We establish the target amount of overlap, and propose a methodology which theoretically guarantees the existence of β^* which makes the posteriors to satisfy the target amount of overlap. On the empirical side, we propose a methodology to find such a value of β^* .

Huggins and Miller (2024) investigate a technique, called *BayesBag*, which makes posteriors more stable under the different data realizations. BayesBag uses bootstraps of data to compute posteriors, and then averages over posteriors from all bootstraps. Formally, Huggins and Miller (2024) compute the bagged posterior as

$$\pi^*(\theta \mid x) = \frac{1}{N^M} \sum_{x^*} \pi(\theta \mid x^*), \quad (6)$$

where $\pi(\theta \mid x^*)$ is the standard posterior of the bootstrap $x^* = (x_1^*, \dots, x_M^*)$, and the sum is taken over all N^M possible bootstrap datasets of size M drawn with replacement from the original dataset $x_{1:N}$. In practice Huggins and Miller (2024) use Monte Carlo estimate of bagged posterior, taking B bootstraps each of size $M = N$.

The contribution of our work is that we not only propose a method for increasing the reproducibility of posterior inference under model misspecification, but also introduce a way to quantify it. We define a threshold that specifies the correct amount of overlap between posteriors computed on independent data realizations of the same DGP. We show that standard Bayes fails to satisfy this threshold under misspecification, while our method can recover it both theoretically and empirically. In the experimental setup, we establish a procedure that finds the value of β^* at which the β -Bayes posteriors achieve the target overlap. As a result, we obtain a notion of stability that is verifiable in practice.

3 Theory and Methodology

In this section we develop the theoretical foundation of our method. In section 3.1 we introduce a way to measure the amount of disagreement between posteriors. Further, in section 3.2, we define the correct amount of overlap between posteriors, which we refer to as *the well-specified threshold*, and introduce *well-calibrated posteriors* via the notion of the *overlap criterion*. Let's note that the well-specified threshold we establish does not depend on the existence of the pseudo-true parameter. Having established the target quantity, in section 3.3 we analyze the expected discrepancy between posteriors in the asymptotic regime, and conclude that under misspecification standard Bayes fails to

satisfy the well-specified threshold. We therefore suggest β -Bayes as a new methodology for recovering well-calibrated posteriors. In section 3.4 we establish sufficient regularity conditions under which β -Bayes posteriors satisfy the well-specified threshold, recovering the correct level of overlap. Finally, in section 3.5, we illustrate that under a variance misspecification in a Gaussian model, the overlap criterion is not satisfied under standard Bayes, but is almost always satisfied under β -Bayes.

3.1 Expected discrepancy

To quantify the amount of overlap between posteriors, we introduce the notion of *expected discrepancy*. Let d be a divergence measure, and let $X, Y \stackrel{\text{iid}}{\sim} P_\circ$ be two independent datasets of size N drawn from the true DGP. Let $\Pi_X = \pi(\theta \mid X)$ and $\Pi_Y = \pi(\theta \mid Y)$ denote the corresponding posteriors. We define the *expected discrepancy* as

$$\bar{d} = \mathbb{E}_{X, Y \sim P_\circ} [d(\Pi_X, \Pi_Y)] = \iint d(\Pi_x, \Pi_y) P_\circ(dx) P_\circ(dy). \quad (7)$$

To proceed we need to choose the divergence measure d to quantify the discrepancy. One of the natural choices is to set d to be the *Rényi divergence* (Rényi, 1961) of order $\alpha = 1/2$,

$$d_{1/2}(\Pi_X, \Pi_Y) = -2 \log \int \sqrt{\Pi_X(\theta) \Pi_Y(\theta)} d\theta = -2 \log \text{BC}(\Pi_X, \Pi_Y), \quad (8)$$

where

$$\text{BC}(\Pi_X, \Pi_Y) = \int \sqrt{\Pi_X(\theta) \Pi_Y(\theta)} d\theta. \quad (9)$$

is the *Bhattacharyya coefficient* (Bhattacharyya, 1943) between the two posteriors. Conceptually, $\text{BC}(\Pi_X, \Pi_Y) \in [0, 1]$ measures the amount of overlap between Π_X and Π_Y — it equals 1 when the two posteriors are identical and 0 when they have completely no overlap. Consequently, $d_{1/2}(\Pi_X, \Pi_Y) \in [0, \infty)$, with $d_{1/2} = 0$ indicating perfect agreement between the two posteriors and larger values indicating greater disagreement. There are

several benefits of using $d_{1/2}$ as the chosen discrepancy measure. It is not very sensitive to differences in tails (as compared to KL and symmetric KL), and is symmetric. Additionally, $d_{1/2}$ is unbounded, having a larger dynamic range, which is an important property for our goal of calibrating posteriors. Moreover, $d_{1/2}$ admits a closed form expression for the pairs of distributions which belong to the same exponential family (Nielsen, Boltz, and Schwander, 2010).

Therefore, we measure the amount of overlap between Π_X and Π_Y as

$$\bar{d}_{1/2} = \iint d_{1/2}(\Pi_x, \Pi_y) P_{\circ}(dx) P_{\circ}(dy). \quad (10)$$

3.2 The well-specified discrepancy threshold

Having defined a measure of disagreement between posteriors, we now define a target discrepancy between posteriors, which represents the correct amount of overlap. When the likelihood is specified correctly, some disagreement between posteriors computed on different data realizations of the same DGP is still expected. Due to the natural sampling variability between the datasets drawn from the same DGP. Therefore, the quantity we target is the amount of disagreement created by differences in datasets, rather than by model misspecification. We refer to this amount as the *well-specified threshold*, and consider it to be the correct level of overlap to target.

Definition 1 (Well-specified threshold). *Assume that the likelihood is correctly specified, meaning that $P_{\circ} = P_{\theta^*}$ for some $\theta^* \in \Theta$ with $\theta^* \sim Q$, where Q is some distribution over Θ . Then, the well-specified threshold τ is the expected discrepancy between posteriors computed on independent datasets drawn from P_{θ^*} :*

$$\tau = \mathbb{E}_{\theta^* \sim Q} \left[\mathbb{E}_{X, Y \sim P_{\theta^*}} [d_{1/2}(\Pi_X, \Pi_Y)] \right].$$

The choice of Q depends on the setting, and can be made by the decision maker. The common choices could be as follows.

1. $Q = \pi(\theta)$ (prior) — the expected overlap under the assumption that the model is

specified correctly, before any data is observed.

2. $Q = \pi(\theta \mid x_{1:N})$ (posterior) — the data-informed expected overlap under the assumption that the model is specified correctly, using the posterior as a belief distribution for θ^* , which would have generated the data.
3. $Q = \delta_\theta$ (point mass) — any $\theta \in \Theta$ can be chosen. Then we generate independent datasets X, Y from P_θ , a member of the likelihood family, and compute the well-specified threshold under the assumed likelihood.

Definition 2 (Overlap criterion; well-calibrated posteriors). *Let $\Pi_X^{(\ell)}, \Pi_Y^{(\ell)}$ denote the GBI posteriors computed on independent datasets $X, Y \sim P_\circ$ under loss ℓ , as defined in (3). The posteriors $\Pi_X^{(\ell)}, \Pi_Y^{(\ell)}$ are said to be well-calibrated or equivalently, to satisfy the overlap criterion, if*

$$\bar{d}_{1/2} = \tau.$$

3.3 Asymptotic discrepancy

The well-specified threshold derived above clearly depends on the assumed likelihood and chosen DGP at finite N . Our goal is to analyze, for a generic model and DGP, whether posteriors under standard Bayes satisfy the well-specified threshold, and if not, to suggest a solution which uses β -Bayes to recover well-calibrated posteriors. Therefore, to have a universal target discrepancy, we analyze the expected discrepancy between posteriors in the asymptotic regime.

Let $\ell(\theta, x)$ be an arbitrary loss function used for GBI as in (3), with corresponding pseudo-true parameter $\theta_*^{(\ell)}$ as defined in (4). We define the information matrices

$$J_{\theta_*^{(\ell)}} = \mathbb{E}[\nabla_\theta^2 \ell(\theta_*^{(\ell)}, x_0)], \quad I_{\theta_*^{(\ell)}} = \text{Cov}_{P_\circ}[\nabla_\theta \ell(\theta_*^{(\ell)}, x_0)],$$

where $x_0 \sim P_\circ$. For brevity we write $J = J_{\theta_*^{(\ell)}}$, $I = I_{\theta_*^{(\ell)}}$. Let X and Y be two independent datasets of size N drawn from $P_\circ$, let $\Pi_X = \pi^{(\ell)}(\theta \mid X)$ and $\Pi_Y = \pi^{(\ell)}(\theta \mid Y)$ be the corresponding GBI posteriors as in (3), and $\bar{\theta}_X, \bar{\theta}_Y$ be the posterior means. We denote

the rescaled posterior means as $\tilde{\theta}_X = \sqrt{N}(\bar{\theta}_X - \theta_*^{(\ell)})$ and $\tilde{\theta}_Y = \sqrt{N}(\bar{\theta}_Y - \theta_*^{(\ell)})$. Using the change of variables $h = \sqrt{N}(\theta - \theta_*^{(\ell)})$, we define the rescaled posteriors as $\tilde{\Pi}_X = \pi^{(\ell)}(h \mid X)$ and $\tilde{\Pi}_Y = \pi^{(\ell)}(h \mid Y)$.

Theorem 1. *If J is nonsingular,*

$$d_{\text{TV}}\left(\tilde{\Pi}_X, \mathcal{N}\left(\tilde{\theta}_X, J^{-1}\right)\right) \xrightarrow{P} 0, \quad (\text{A1.1})$$

$$d_{\text{TV}}\left(\mathcal{L}\left(\tilde{\theta}_X\right), \mathcal{N}\left(0, J^{-1} I J^{-1}\right)\right) \xrightarrow{P} 0, \quad (\text{A1.2})$$

the same conditions hold for Y , and

$$d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y) \text{ is uniformly integrable,} \quad (\text{A1.3})$$

then

$$\bar{d}_{1/2} = \mathbb{E}[d_{1/2}(\Pi_X, \Pi_Y)] \rightarrow \frac{1}{2} \text{Tr}(I J^{-1}). \quad (11)$$

The assumptions (A1.1), (A1.2) are in fact the Bernstein-von Mises theorem (BvM) and central limit theorem for the MLE under misspecification. Define the Gaussian approximations to $\tilde{\Pi}_X, \tilde{\Pi}_Y$ respectively as $\phi_X = \mathcal{N}(\tilde{\theta}_X, J^{-1})$ and $\phi_Y = \mathcal{N}(\tilde{\theta}_Y, J^{-1})$. Before proving Theorem 1, we state the lemma, required for the proof.

Lemma 1. *Under assumptions (A1.1) and (A1.3),*

$$\left| \mathbb{E}[d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y)] - \mathbb{E}[d_{1/2}(\phi_X, \phi_Y)] \right| \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

The proof is given in Appendix A.

The divergence $d_{1/2}$ is invariant to reparametrization (Erven and Harremoës, 2014), which gives

$$d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y) = d_{1/2}(\Pi_X, \Pi_Y).$$

Proof of Theorem 1. Lemma 1 can be applied, thus, it suffices to show that (11) holds for $\mathbb{E}[d_{1/2}(\phi_X, \phi_Y)]$.

For two Gaussian distributions with the same covariance J^{-1} and means $\tilde{\theta}_X$ and $\tilde{\theta}_Y$, the Bhattacharyya coefficient admits the closed form (Nielsen, Boltz, and Schwander, 2010),

$$\text{BC}(\phi_X, \phi_Y) = \int \sqrt{\phi_X(\theta)\phi_Y(\theta)} d\theta = \exp\left(-\frac{1}{8}(\tilde{\theta}_X - \tilde{\theta}_Y)^\top J (\tilde{\theta}_X - \tilde{\theta}_Y)\right). \quad (12)$$

As $d_{1/2} = -2 \log \text{BC}$, we get

$$d_{1/2}(\phi_X, \phi_Y) = \frac{1}{4}(\tilde{\theta}_X - \tilde{\theta}_Y)^\top J (\tilde{\theta}_X - \tilde{\theta}_Y). \quad (13)$$

By assumption (A1.2), the rescaled posterior means satisfy

$$\tilde{\theta}_X \xrightarrow{d} \mathcal{N}(0, J^{-1}IJ^{-1}), \quad \tilde{\theta}_Y \xrightarrow{d} \mathcal{N}(0, J^{-1}IJ^{-1}).$$

Therefore, their difference satisfies

$$\tilde{\theta}_X - \tilde{\theta}_Y \xrightarrow{d} \mathcal{N}(0, 2J^{-1}IJ^{-1}).$$

Therefore, using $\mathbb{E}[z^\top Az] = \text{Tr}(A\Sigma)$ for $z \sim \mathcal{N}(0, \Sigma)$ with $A = \frac{1}{4}J$ and $\Sigma = 2J^{-1}IJ^{-1}$, we get

$$\begin{aligned} \mathbb{E}[d_{1/2}(\Pi_X, \Pi_Y)] &\rightarrow \mathbb{E}[d_{1/2}(\phi_X, \phi_Y)] = \\ &= \frac{1}{4} \mathbb{E}[(\tilde{\theta}_X - \tilde{\theta}_Y)^\top J (\tilde{\theta}_X - \tilde{\theta}_Y)] \\ &= \frac{1}{4} \text{Tr}(J \cdot 2J^{-1}IJ^{-1}) \\ &= \frac{1}{2} \text{Tr}(IJ^{-1}). \end{aligned}$$

□

Remark. Miller (2021) states the necessary conditions under which the assumption (A1.1) holds for generalized posteriors. The conditions on the loss function include continuous third derivatives, a positive definite Hessian at $\theta_*^{(\ell)}$, uniformly bounded third derivatives, and identifiability. For the β -divergence loss with a finite $\beta > 1$, these conditions are typically satisfied for smooth likelihood families.

Under the likelihood being correctly specified we have $I = J$. Therefore, the result in (11) simplifies to $\frac{1}{2}\text{Tr}(IJ^{-1}) = \frac{1}{2}\text{Tr}(I_D) = \frac{D}{2}$, where D is the dimension of parameter space Θ . Therefore, asymptotically, the well-specified threshold is

$$\tau = \frac{D}{2}.$$

Clearly, under standard Bayes, the asymptotic expected discrepancy $\frac{1}{2}\text{Tr}(IJ^{-1})$ depends only on the parameters of the true DGP and the selected model, meaning that under misspecification no parameters can be calibrated in order to make the expected discrepancy to satisfy the well-specified threshold. Under β -Bayes, the parameter β is included into the objective (11) and can be calibrated in order to change the expected discrepancy to match the well-specified threshold $\frac{D}{2}$.

3.4 Expected discrepancy under β -Bayes

Above we determined the universal value of the well-specified threshold in the asymptotic regime. We further establish a set of sufficient conditions under which, for any finite upper bound $C < \infty$, there exists $\beta^* \in (1, C]$ such that the β -Bayes posteriors at β^* are well-calibrated. We consider two regimes:

1. Above-threshold regime: standard Bayes asymptotically gives a discrepancy higher than $\frac{D}{2}$ (meaning that the posteriors overlap less than they would under the correct model).
2. Below-threshold regime: standard Bayes asymptotically gives a discrepancy lower than $\frac{D}{2}$ (meaning that the posteriors overlap more than they would under the correct model).

Intuitively, β -Bayes should in most cases be able to improve the expected discrepancy in above-threshold regime, but not in below-threshold regime, as it is supposed to robustify the posterior and make it to depend less on the wrong model. In below-threshold regime it is already the case that posteriors are overlapping too much, therefore, β -Bayes in most cases will only make the overlap bigger, rather than decreasing it.

Let $\ell^{(\beta)}$ be the β -divergence loss as defined in (5), and let $I_{\theta_*^{(\beta)}}$ and $J_{\theta_*^{(\beta)}}$ be as defined in Theorem 1, with the exact forms given in Appendix C.

Theorem 2. *Let $C > 1$ be an arbitrary fixed constant. Suppose in above-threshold regime the conditions of Theorem 1 are satisfied for all $\beta \in (1, C]$, and the following assumptions hold:*

$$p(x|\theta_*^{(\beta)}) > 0 \quad \text{for } P_\circ\text{-almost all } x, \quad (\text{A2.1})$$

For every $\beta_0 \in (1, C]$, let $\delta = \delta(\beta_0) > 0$ be such that $B(\theta_*^{\beta_0}, \delta) \subset \Theta$. Then, for every $\beta_0 \in (1, C]$,

$$\sup_{\theta \in B(\theta_*^{\beta_0}, \delta)} \mathbb{E}_{P_\circ} [\max(1, p(x|\theta)^{2(C-1)}) \|\nabla_\theta \log p(x|\theta)\|^2] < \infty, \quad (\text{A2.2})$$

$$\sup_{\theta \in B(\theta_*^{\beta_0}, \delta)} \mathbb{E}_{P_\circ} [\max(1, p(x|\theta)^{C-1}) (\|\nabla_\theta^2 \log p(x|\theta)\| + \|\nabla_\theta \log p(x|\theta)\|^2)] < \infty, \quad (\text{A2.3})$$

$$\sup_{\theta \in B(\theta_*^{\beta_0}, \delta)} \int \max(p(z|\theta), p(z|\theta)^C) (\|\nabla_\theta^2 \log p(z|\theta)\| + \|\nabla_\theta \log p(z|\theta)\|^2) dz < \infty, \quad (\text{A2.4})$$

$$\beta \mapsto \theta_*^{(\beta)} \text{ is continuous on } (1, C], \quad (\text{A2.5})$$

$$\exists \beta' \in (1, C] \text{ such that } \frac{1}{2} \text{Tr} \left(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1} \right) \Big|_{\beta=\beta'} < \frac{D}{2}. \quad (\text{A2.6})$$

Then, there exists a value of $\beta^* \in (1, C]$ such that the β -Bayes posterior with β^* is asymptotically well-calibrated, meaning that

$$\bar{d}_{1/2} \rightarrow \tau \quad \text{as } N \rightarrow \infty.$$

Before proving Theorem 2, we introduce a necessary Lemma.

Lemma 2. *Under assumptions (A2.1)–(A2.5), $\text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1})$ is a continuous function of β on $(1, C]$. The proof is given in Appendix B.*

Proof of Theorem 2. By Lemma 2, $\text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1})$ is continuous in β for all $\beta \in (1, C]$. In above-threshold regime, $\frac{1}{2}\text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1})|_{\beta \rightarrow 1} > \frac{D}{2}$, while by assumption (A2.6) there exists a finite $\beta' > 1$ such that $\frac{1}{2}\text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1})|_{\beta=\beta'} < \frac{D}{2}$. Therefore, by the Intermediate Value Theorem, there exists $\beta^* \in (1, \beta')$ such that $\frac{1}{2}\text{Tr}(I_{\theta_*^{(\beta^*)}} J_{\theta_*^{(\beta^*)}}^{-1}) = \frac{D}{2}$. By Theorem 1, this implies $\bar{d}_{1/2} \rightarrow \tau$, so the β^* -Bayes posterior is asymptotically well-calibrated. $\square$

Theorem 3. *Suppose in below-threshold regime the conditions of Theorem 1 are satisfied, assumptions (A2.1) – (A2.5) hold, and additionally:*

$$\exists \beta' \in (1, C] \text{ such that } \frac{1}{2}\text{Tr}\left(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1}\right)\Big|_{\beta=\beta'} > \frac{D}{2}. \quad (\text{A2.7})$$

Then, there exists a value of β^* such that β -Bayes posterior with β^* is asymptotically well-calibrated, meaning that

$$\bar{d}_{1/2} \rightarrow \tau.$$

Proof of Theorem 3. The proof is analogous to the proof of the Theorem 2, replacing assumption (A2.6) with (A2.7). $\square$

The analysis in the asymptotic regime shows that in principle β -Bayes can be helpful for recovering well-calibrated posteriors. Further in section 4 we empirically show that the methodology can be used in practice.

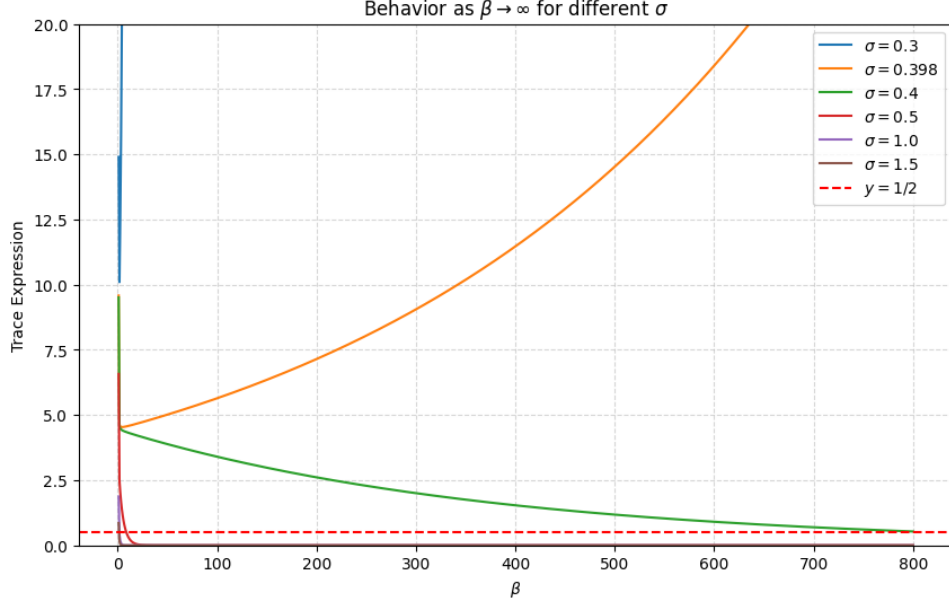


Figure 2: Behavior of $\frac{1}{2} \text{Tr}(I_{\theta_o(\beta)} J_{\theta_o(\beta)}^{-1})$ for different values of σ and fixed $s = 2.0$, as $\beta \rightarrow \infty$, in the above-threshold regime.

3.5 Non-asymptotic analysis

Below we apply the conditions derived in Theorem 2 and Theorem 3, to an example with a misspecified model. We consider a case of a simple Gaussian location model with a misspecified variance. We want to check whether in any of the regimes there exists a value of β which will allow the asymptotic expected discrepancy to be equal to $1/2$, which is the well-specified threshold in $D = 1$.

The true DGP is $P_o = \mathcal{N}(\mu^*, s^2)$ and the likelihood is $p(x | \theta) = \mathcal{N}(x | \theta, \sigma^2)$ with misspecified variance $\sigma^2 \neq s^2$. Given the setup, we derive that

$$\frac{1}{2} \text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1}) = \frac{(2\pi\sigma^2)^{-(\beta-1)/2} s^2}{2\sigma^2} \left(\frac{1 + \frac{(\beta-1)s^2}{\sigma^2}}{1 + \frac{2(\beta-1)s^2}{\sigma^2}} \right)^{3/2}. \quad (14)$$

Clearly, the objective in (14) is continuous in β . Therefore, the conditions (A2.1) – (A2.4) are automatically satisfied. Additionally, we can derive that $\theta_*^{(\beta)} = \mu^*$, and therefore does not change with β , meaning that the condition (A2.5) is also automatically satisfied. Below we check conditions (A2.6) and (A2.7) for above-threshold and below-threshold regimes respectively.

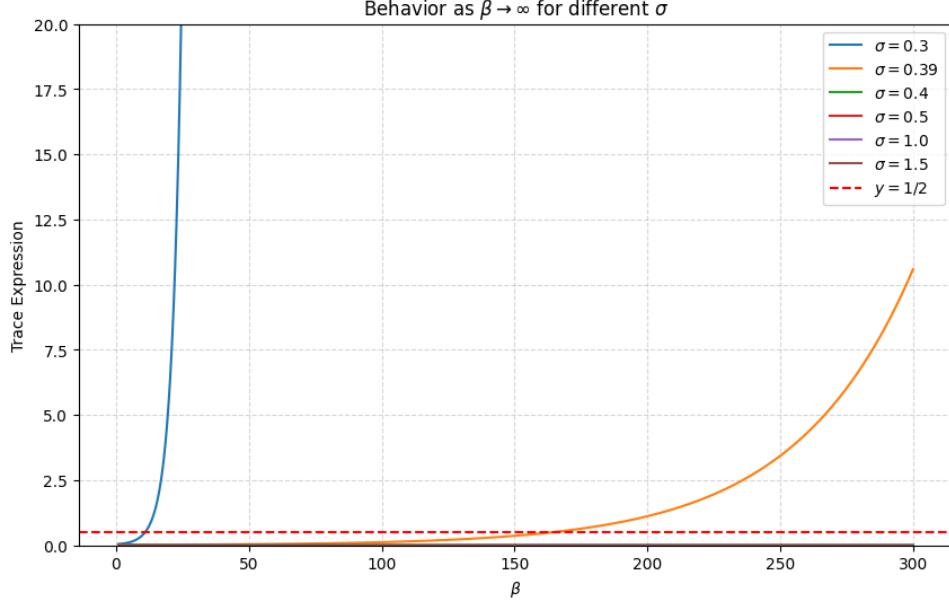


Figure 3: Behavior of $\frac{1}{2} \text{Tr}(I_{\theta_{\circ}(\beta)} J_{\theta_{\circ}(\beta)}^{-1})$ for different values of σ and fixed $s = 0.1$, as $\beta \rightarrow \infty$, in the below-threshold regime.

Let's denote that for standard Bayes, the asymptotic expected discrepancy is

$$\frac{1}{2} \text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1}) \Big|_{\beta \rightarrow 1} = \frac{s^2}{2\sigma^2}.$$

Above-threshold regime. In this regime, the asymptotic expected discrepancy under standard Bayes exceeds the well specified threshold, meaning that

$$\frac{s^2}{\sigma^2} > 1.$$

Clearly, the objective (14) is a monotonically decreasing function of β if $\sigma^2 > \frac{1}{2\pi} \approx 0.16$, in the units in which σ^2 is expressed. Thus, there exists a value β' such that $\frac{1}{2} \text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1}) \Big|_{\beta=\beta'} < \frac{1}{2}$. Therefore, if the model variance is greater than 0.16, by Theorem 2, there exists a value of β^* such that $\frac{1}{2} \text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1}) \Big|_{\beta=\beta^*} = \frac{1}{2}$. On figure 2 we can see that for values of $\sigma < \sqrt{2\pi}$, the objective does not intersect the value of $\frac{1}{2}$, while for values above it does.

Below-threshold regime. In below-threshold regime, $\frac{s^2}{\sigma^2} < 1$, and the posteriors under the misspecified model already overlap more than they should. Under the same condition

$\sigma^2 > \frac{1}{2\pi}$, the objective (14) is monotonically decreasing in β . Since the objective starts below $\frac{1}{2}$ at $\beta \rightarrow 1$ and decreases further, it never crosses $\frac{1}{2}$. On figure 3, we can see that for value of $\sigma > \sqrt{2\pi}$, the objective does not cross the threshold. Therefore, Theorem 3 does not guarantee the existence of $\beta^* > 1$. Intuitively, such β^* cannot exist, as β -Bayes downweights the likelihood, making the posterior even less confident, which only increases the overlap further.

4 Experiments

In this section we consider several experimental setups, where the assumed likelihood is misspecified. We conduct experiments both with synthetic and real data. The empirical well-specified threshold is estimated using different distributions Q , introduced in 1. In the case of data being synthetically generated, Q is chosen to be the point mass $Q = \delta_\theta$. In experiments with real data, we set Q to be the posterior $\pi(\theta \mid x_{1:N})$.

In each of the experiments our goal is to establish the value of the well-specified threshold, of the observed discrepancy under standard Bayes, and of the observed discrepancy under β -Bayes for some value of β . Then, we proceed by finding the value of β^* for which the observed discrepancy under β -Bayes matches the well-specified threshold.

Additionally, for real data experiments, we analyze the effect on downstream performance. We investigate stability of parameter inference by comparing mean pairwise L_2 distances between posterior means and mean pairwise Frobenius norms between posterior covariances across blocks of data. We investigate stability of predictive inference by comparing mean pairwise $d_{1/2}$ distances between posterior predictives across blocks of data, evaluated on a held-out test set. Finally, to assess the effect on predictive accuracy, we compute such metrics as Mean Log predictive Density (MLPD), Continuous Ranked Probability Score (CRPS), Root Mean Squared Error (RMSE), and predictive coverage.

4.1 Synthetic data: Gaussian location model with misspecified variance

Data In this example the goal of the posterior inference is the mean parameter μ . We generate the data from the normal distribution $\mathcal{N}(\mu^*, s^2)$, where we set $\mu^* = 0.0$ and $s^2 = 1.5$. The prior is $\mathcal{N}(\mu_0, \tau_0^2)$, with $\mu_0 = 0$ and $\tau_0^2 = 0.5$. We deliberately set the variance $\sigma^2 = 0.5$ to be misspecified.

Our goal is to compute the well-specified threshold discrepancy, the observed discrepancy under standard Bayes and under β -Bayes, and to find the value of β^* for which the observed discrepancy is close to the well-specified threshold.

Well-specified threshold By Definition 1, the well-specified threshold is computed under the assumption that the likelihood is correctly specified. Therefore, to estimate it, we generate the data from the likelihood, choosing $Q = \delta_{\mu^*}$ for an arbitrary $\mu^* \in \mathbb{R}$. Note that the choice of μ^* does not affect the value of the well-specified threshold, as it only affects the mean location, rather than the amount of overlap.

Let $X_{1i}, X_{2i} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu^*, \sigma^2)$ for $i = 1, \dots, M$ be the pairs of independent datasets, drawn from the assumed likelihood. We define the well-specified posteriors as the posteriors computed under the assumed likelihood:

$$\pi_{ji}^* := \pi(\mu \mid X_{ji}, \sigma^2), \quad j = 1, 2, \quad i = 1, \dots, M.$$

We estimate the well-specified threshold as

$$\begin{aligned} d_{1/2}^* &= \frac{1}{M} \sum_{i=1}^M d_{1/2}(\pi_{1i}^*, \pi_{2i}^*) \\ &= \frac{1}{M} \sum_{i=1}^M -2 \log \left(\int \sqrt{\pi_{1i}^*(\mu) \pi_{2i}^*(\mu)} d\mu \right). \end{aligned}$$

In this set up, the well-specified posteriors are Gaussian, with means

$$\mu_{ji} = \tau_n^2 \left(\frac{n \bar{X}_{ji}}{\sigma^2} + \frac{\mu_0}{\tau_0^2} \right), \quad j = 1, 2,$$

and shared posterior variance

$$\tau_n^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\tau_0^2} \right)^{-1}.$$

The well-specified threshold simplifies to

$$d_{1/2}^* = \frac{1}{M} \sum_{i=1}^M \frac{(\mu_{1i} - \mu_{2i})^2}{4\tau_n^2}.$$

Observed discrepancy under standard Bayes Analogously, we define the observed discrepancy under standard Bayes as

$$\hat{d}_{1/2} = \frac{1}{M} \sum_{i=1}^M d_{1/2}(\hat{\pi}_{1i}, \hat{\pi}_{2i}),$$

where $\hat{\pi}_{ji} := \pi(\mu \mid X_{ji}, \sigma^2)$ is the posterior computed under the misspecified likelihood with variance $\sigma^2 \neq s^2$. The observed discrepancy simplifies to

$$\hat{d}_{1/2} = \frac{1}{M} \sum_{i=1}^M \frac{(\hat{\mu}_{1i} - \hat{\mu}_{2i})^2}{4\hat{\tau}_n^2}.$$

Observed discrepancy under β -Bayes For β -Bayes, the posterior is no longer Gaussian, and no closed form expression for the posterior is available. Therefore, we evaluate the Bhattacharyya coefficient numerically using a grid of values $\{\mu_k\}_{k=1}^K$ on $[\mu_{\min}, \mu_{\max}]$:

$$BC(\tilde{\pi}_{1i}, \tilde{\pi}_{2i}) \approx \sum_k \sqrt{\tilde{\pi}_{1i}(\mu_k) \cdot \tilde{\pi}_{2i}(\mu_k)} \cdot \Delta\mu,$$

which corresponds to applying the trapezoid rule to $\int \sqrt{\tilde{\pi}_{1i}(\mu) \tilde{\pi}_{2i}(\mu)} d\mu$. Then, the discrepancy under generalized Bayes is

$$\tilde{d}_{1/2}(\beta) = \frac{1}{M} \sum_{i=1}^M -2 \log \left(BC(\tilde{\pi}_{1i}(\beta), \tilde{\pi}_{2i}(\beta)) \right).$$

Experimental conditions We use dataset size $n = 1000$ and $M = 1000$ independent dataset pairs for Monte Carlo averaging. We reuse the same datasets drawn from the true

DGP for computation of all thresholds. For numerical evaluation of β -Bayes posteriors and BC integrals, we use a grid of $K = 5000$ points on $[\mu_{\min}, \mu_{\max}] = [-2, 2]$.

Finding β^* Our goal is to establish the value of β^* such that

$$\beta^* = \arg \min_{\beta > 1} \left| \tilde{d}_{1/2}(\beta) - d_{1/2}^* \right|,$$

or equivalently, find the root of

$$\tilde{d}_{1/2}(\beta) - d_{1/2}^* = 0.$$

In practice, we solve this using a root-finding algorithm over a grid of β values.

Results As expected, and shown in the asymptotic regime in section 3.5, the observed threshold under standard Bayes in this misspecification example is larger than the well-specified threshold. Following the methodology described above, we find the value of β^* for which the observed discrepancy under β -Bayes matches the well-specified threshold. We summarize the computed discrepancies (MC averages) and their standard deviations in the table below.

We can see that β -Bayes not only recovers the well-specified threshold, but also has a variability of the pairwise posterior discrepancy $d_{1/2}$ across dataset pairs similar to that of the well-specified model.

Additionally, let's denote that the value of the well-specified threshold $d_{1/2}^* = 0.4817$ is close to the well-specified threshold in the asymptotic regime, $\tau = \frac{1}{2}$.

	Mean	Std
$d_{1/2}^*$	0.4963	0.700
$\hat{d}_{1/2}$	1.4481	2.105
$\tilde{d}_{1/2}(\beta^*), \beta^* = 1.6214$	0.4963	0.669

Table 1: Pairwise posterior discrepancies $d_{1/2}$ across $M = 1000$ independent dataset pairs. The Mean column reports the average over the M pairs, defining $d_{1/2}^*$, $\hat{d}_{1/2}$, and $\tilde{d}_{1/2}(\beta^*)$. The Std column reports the standard deviation of the per-pair values across the M pairs. $\beta^* = 1.6492$ is chosen so that $\tilde{d}_{1/2}(\beta^*) = d_{1/2}^*$.

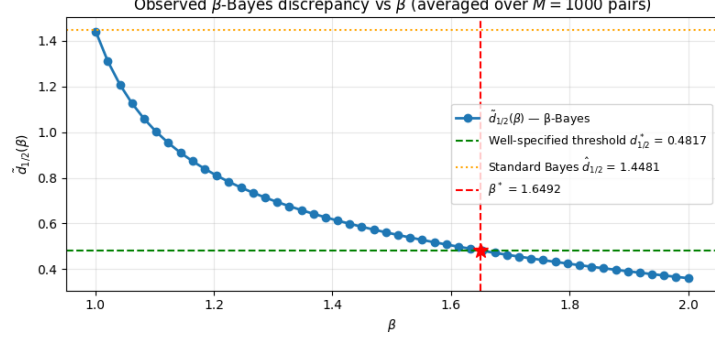


Figure 4: Observed discrepancy under β -Bayes, $\tilde{d}_{1/2}(\beta)$ as a function of β . The well-specified threshold – $d_{1/2}^* = 0.4817$ is shown in green and observed discrepancy under standard Bayes $\hat{d}_{1/2} = 1.4481$ is shown in orange. The value of $\beta^* = 1.6492$ is the value for which $\tilde{d}_{1/2}(\beta^*) = d_{1/2}^*$.

Figure 4 shows how the value of $\tilde{d}_{1/2}(\beta)$ changes as a function of β on the figure below. As expected by the results of asymptotic example, $\tilde{d}_{1/2}(\beta)$ is close to $\hat{d}_{1/2}$ for values of β close to 1, and monotonically decreases, crossing the well specified threshold $\hat{d}_{1/2}$ for $\beta^* = 1.6492 > 1$.

4.2 Synthetic data: regression with heavy noise

Data In this example we generate synthetic data from a linear regression model with heavy-tailed noise. Specifically, the true data-generating process is

$$y_i = \mathbf{x}_i^\top \boldsymbol{\delta}^* + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} t(5), \quad i = 1, \dots, n,$$

where $\mathbf{x}_i \in \mathbb{R}^D$ with $D = 4$ are uncorrelated standard Gaussian features, and the true coefficient vector is

$$\boldsymbol{\delta}^* = \left(\frac{4}{\sqrt{1}}, \frac{4}{\sqrt{2}}, \frac{4}{\sqrt{3}}, \frac{4}{\sqrt{4}} \right)^\top = (4.00, 2.83, 2.31, 2.00)^\top.$$

The assumed likelihood is a Gaussian linear regression model, which is misspecified since the true noise is $t(5)$. We place a Normal-Inverse-Gamma (NIG) prior on the regression

coefficients $\boldsymbol{\delta}$ and the noise variance σ^2 , which is conjugate to the Gaussian likelihood:

$$\begin{aligned}\sigma^2 &\sim \mathcal{IG}(a_0, b_0), \\ \boldsymbol{\delta} \mid \sigma^2 &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{\Sigma}_0),\end{aligned}$$

where $\mathcal{IG}(a_0, b_0)$ denotes the inverse-gamma distribution with shape a_0 and scale b_0 , and $\boldsymbol{\Sigma}_0$ is the prior covariance matrix. The prior hyperparameters are set to $a_0 = 0.5$, $b_0 = 0.5$, and $\boldsymbol{\Sigma}_0 = 2.0 \cdot \mathbf{I}_D$.

Well-specified threshold By Definition 1, the well-specified threshold is computed under the assumption that the likelihood is correctly specified. Therefore, we generate the data from the assumed Gaussian likelihood, choosing $Q = \delta_{\theta^*}$ with $\theta^* = (\boldsymbol{\delta}^*, \sigma^{*2})$, where $\boldsymbol{\delta}^*$ is the true regression coefficient vector and $\sigma^{*2} = \text{Var}(t(5)) = 5/3$ is chosen to match the variance of the true noise distribution. We simulate the data from the assumed model as

$$y_{ji} = \mathbf{X}_{ji} \boldsymbol{\delta}^* + \varepsilon_{ji}, \quad \varepsilon_{ji} \sim \mathcal{N}(0, \sigma^{*2} \mathbf{I}_n), \quad j = 1, 2, \quad i = 1, \dots, M.$$

Under the assumed Gaussian likelihood and NIG prior, the well-specified posteriors are NIG and available in closed form: $\pi_{ji}^* = \text{NIG}(\boldsymbol{\mu}_{ji}, \mathbf{V}_{ji}, a_n, b_{ji})$, with the parameters given in Appendix C. For NIG π_{1i}^*, π_{2i}^* , the well-specified threshold

$$d_{1/2}^* = \frac{1}{M} \sum_{i=1}^M -2 \log BC(\pi_{1i}^*, \pi_{2i}^*)$$

can be computed in closed form, which is stated in Appendix C.

Observed discrepancy under standard Bayes The observed discrepancy is computed on M pairs of datasets generated from the true DGP. The NIG posteriors $\hat{\pi}_{ji}$ are computed under the assumed Gaussian likelihood. The observed discrepancy, which we

define as

$$\hat{d}_{1/2} = \frac{1}{M} \sum_{i=1}^M -2 \log BC(\hat{\pi}_{1i}, \hat{\pi}_{2i}),$$

is computed analogously to the well-specified threshold.

Observed discrepancy under β -Bayes Under β -Bayes, the generalized posterior is not available in closed form. We use a method proposed by Knoblauch, Jewson, and Damoulas (2022), Generalized Variational Inference (GVI), in order to approximate the posterior under β -divergence loss. GVI allows to find the best approximate posterior within a chosen variational family $\mathcal{Q}$, by solving an optimization problem

$$q^*(\boldsymbol{\delta}, \sigma^2) = \arg \min_{q \in \mathcal{Q}} \mathbb{E}_{q(\boldsymbol{\delta}, \sigma^2)} \left[\sum_{i=1}^n \ell^{(\beta)}(y_i, \mathbf{x}_i, \boldsymbol{\delta}, \sigma^2) \right] + \text{KL}(q(\boldsymbol{\delta}, \sigma^2) \parallel \pi(\boldsymbol{\delta}, \sigma^2)).$$

We use the NIG family as $\mathcal{Q}$, and initialize the variational parameters at the standard NIG posterior. Since the resulting approximate posterior q^* is NIG, the same closed form expression for the BC as above can be applied, and the observed discrepancy $\tilde{d}_{1/2}(\beta)$ can be computed analytically.

Experimental conditions We use dataset size $n = 1000$ and $M = 1000$ independent dataset pairs for Monte Carlo averaging.

Results Following the methodology described above, we find the value of β^* for which the observed discrepancy under β -Bayes matches the well-specified threshold. We summarize the computed discrepancies (MC averages) and their standard deviations in the table 2.

We can see that β -Bayes not only recovers the well-specified threshold, but also has a variability of the pairwise posterior discrepancy $d_{1/2}$ across dataset pairs similar to that of the well-specified model.

4.3 Real data: California Housing Dataset

Data The California housing dataset (Nugent, 2017) contains $n = 20,640$ observations, each corresponding to a census block group in California. The dataset has $D = 8$ features, including median income, house age, average number of rooms and bedrooms, population, average occupancy, and geographic coordinates (latitude and longitude). The target variable is the median house value for each block group, expressed in units of \$100,000. We are using the Gaussian linear regression likelihood

$$y_i = \mathbf{x}_i^\top \boldsymbol{\delta}^* + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n.$$

We are using the same NIG prior as in the previous example.

Well-specified threshold In this case, we use the posterior option of Q , as stated in Definition 1. We define the well-specified threshold as the expected posterior discrepancy under the assumption that the Gaussian likelihood is correctly specified, using the parameters drawn from standard posterior.

We first fit standard NIG posterior to the full dataset, obtaining a posterior $\hat{\pi}(\boldsymbol{\delta}, \sigma^2 \mid X, \mathbf{y})$. Then we draw M pairs of parameter vectors, $(\boldsymbol{\delta}_i, \sigma_i^2) \sim \hat{\pi}$ ($i \in \{1, 2, \dots, M\}$), and simulate two independent datasets for each pair from the assumed Gaussian likelihood based on these parameters:

$$\mathbf{y}_{ji} = X_{\text{block}} \boldsymbol{\delta}_i + \boldsymbol{\varepsilon}_{ji}, \quad \boldsymbol{\varepsilon}_{ji} \sim \mathcal{N}(\mathbf{0}, \sigma_i^2 \mathbf{I}), \quad j = 1, 2,$$

	Mean	Std
Well-specified threshold $d_{1/2}^*$	2.567	1.581
Standard Bayes $\hat{d}_{1/2}$	4.965	6.704
β -Bayes $\tilde{d}_{1/2}(\beta^*)$, $\beta^* = 1.05$	2.567	1.716

Table 2: Pairwise posterior discrepancies $d_{1/2}$ across $M = 1000$ independent dataset pairs. The Mean column reports the average over the M pairs, defining $d_{1/2}^*$, $\hat{d}_{1/2}$, and $\tilde{d}_{1/2}(\beta^*)$. The Std column reports the standard deviation of the per-pair values across the M pairs. $\beta^* = 1.05065$ is chosen so that $\tilde{d}_{1/2}(\beta^*) = d_{1/2}^*$.

where X_{block} is a subset of X of size n/K , chosen of this size in order to be comparable to observed discrepancies. We then fit the standard NIG posterior to each simulated dataset and compute BC between the two resulting posteriors in closed form, as the posteriors are NIG. The well-specified threshold is defined as the average discrepancy over M replicates:

$$d_{1/2}^* = \frac{1}{M} \sum_{m=1}^M -2 \log BC(\hat{\pi}_{1i}, \hat{\pi}_{2i}). \quad (15)$$

Observed discrepancy under standard Bayes In the example of real-world data we are unable to synthetically generate many independent datasets. We are given a single dataset, and we need to derive the empirical approximation of $d_{1/2}$ based on it. For this purpose, we split the dataset into K blocks of equal size, fit the standard NIG posterior to each block, and compute the U -statistic of the observed pairwise discrepancies. This gives an unbiased estimator of the discrepancy,

$$\hat{d}_{1/2} = \frac{2}{K(K-1)} \sum_{j=1}^K \sum_{k=j+1}^K d_{1/2}(\hat{\pi}_j, \hat{\pi}_k) = \frac{2}{K(K-1)} \sum_{j=1}^K \sum_{k=j+1}^K -2 \log BC(\hat{\pi}_j, \hat{\pi}_k), \quad (16)$$

where $BC(\hat{\pi}_j, \hat{\pi}_k)$ can be computed in closed form as the posteriors are NIG.

Observed discrepancy under β -Bayes In order to approximate the posterior, which is not available in closed form under β -Bayes, we are using GVI with the NIG variational family, as in the previous example. Approximated posteriors are NIG, therefore, in order to approximate observed discrepancy under β -Bayes, we are using the same method as for the observed discrepancy under standard Bayes.

Experimental conditions We use $M = 1000$ and $K = 8$.

Results In this real data example, applying the methodology described above, we were able to establish the value of β^* , which recovers well-calibrated posteriors.

Downstream performance We have established a way to recover well-calibrated posteriors under model misspecification, using β -Bayes. In the real-world setting, researchers

	Mean
Well-specified threshold $d_{1/2}^*$	4.5728
Standard Bayes $\hat{d}_{1/2}$	74.0196
β -Bayes $\tilde{d}_{1/2}(\beta^*)$, $\beta^* = 1.1942$	4.5729

Table 3: Mean pairwise posterior discrepancies $d_{1/2}$ across $M = 1000$ independent dataset pairs. $\beta^* = 1.1942$ is chosen so that $\tilde{d}_{1/2}(\beta^*) = d_{1/2}^*$.

are focused on such downstream tasks as parameter inference and predictive inference. Therefore, we investigate whether the proposed methodology positively affects any of these.

On the California Housing dataset, we conduct several measurements in order to establish whether using β -Bayes improves stability of parameter inference and both stability and accuracy of predictive inference.

We establish that parameter inference is significantly more stable under β -Bayes than under standard Bayes. We compute posteriors on 8 blocks of the data, and the posterior mean vector and covariance matrix for each block. To assess stability across data splits, we compute the $L2$ distance between the mean vectors from different blocks, and the Frobenius norm between the covariance matrices from different blocks. For standard Bayes and several β values we get the results provided in the Table 4. Therefore, we can

	Standard Bayes	$\beta^* = 1.194$	$\beta = 1.1$	$\beta = 1.2$	$\beta = 1.3$	$\beta = 1.5$
Mean L2 distance	1.3661	0.2545	1.2255	0.2535	0.2485	0.2555
Mean Frobenius norm	0.0340	0.0257	0.0520	0.0260	0.0282	0.0299

Table 4: Mean L2 distance between posterior means and Frobenius norm between posterior covariances across blocks for standard Bayes and β -Bayes at various β values.

conclude that β -Bayes significantly improves the stability to the choice of data realizations, as compared to standard Bayes. Additionally, we can notice that the quantities are not significantly sensitive to the choice of moderate value of β .

We next investigate whether a significant increase in stability of posteriors for β -Bayes leads to an increase in stability of posterior predictives. Fixing a testing set of points, we fit the posterior for each of the 8 blocks and compute pairwise discrepancies for standard Bayes and several values of β . In table 5 we can see that the mean pairwise discrepancy

between posterior predictives is significantly improved under β -Bayes.

Model	$d_{1/2}(\beta)$
Standard Bayes	8.5717
$\beta=1.1$	8.1025
β^*	0.5707
$\beta=1.3$	0.6331
$\beta=1.5$	0.6478

Table 5: Mean $d_{1/2}(\beta)$ between posterior predictives for each model.

We explore additional metrics, such as MLPD, CRPS, RMSE, and predictive coverage, which allow us to understand whether the use of β -Bayes improves stability and accuracy of posterior predictives. The mean values and standard errors for these metrics for all the blocks are provided in table 6. The results in this example illustrate that β -Bayes

Model	MLPD	RMSE	Cov90	Cov95	Cov99	CRPS
Standard Bayes	-0.9532	0.6286	0.917	0.944	0.971	0.3369
(SE)	0.0071	0.0045	0.002	0.002	0.001	0.0019
$\beta=1.1$	-1.0745	1.8421	0.904	0.934	0.966	0.3512
(SE)	0.0602	0.6387	0.002	0.002	0.001	0.0158
β^*	-1.0625	1.7936	0.896	0.928	0.962	0.3498
(SE)	0.0532	0.6161	0.002	0.002	0.002	0.0154
$\beta=1.3$	-1.0500	1.7195	0.890	0.924	0.960	0.3484
(SE)	0.0463	0.5801	0.003	0.002	0.002	0.0147
$\beta=1.5$	-1.0186	1.6372	0.892	0.925	0.960	0.3475
(SE)	0.0360	0.5383	0.003	0.002	0.002	0.0138

Table 6: Predictive performance metrics for each model.

does not improve accuracy of predictions as compared to standard Bayes.

Therefore, in this example, we can conclude that β -Bayes helps to significantly improve stability of parameter inference and predictive inference, but not the accuracy of predictive inference.

Based on this example, we can conclude that targeting the discrepancy between posteriors using β -Bayes is justified when the goal is stable parameter inference or predictive inference. When the goal is an accurate point estimation, β -Bayes does not show improvement over standard Bayes. Additionally, we can observe that the stability metrics

seem to be not particularly sensitive to the choice of β , when it is reasonably far from standard Bayes ($\beta = 1.0$).

4.4 Real data: Appliances Energy Prediction Dataset

Data Appliances Energy Prediction dataset (Candanedo, 2017) contains 19,735 observations of household energy consumption collected at 10-minute intervals over approximately 4.5 months in a low-energy building in Belgium. The target variable is the energy use of appliances, and the 28 features include temperature and humidity readings from nine different rooms in the house, as well as weather measurements. We remove features with pairwise correlation greater than 0.95, resulting in 24 features.

The inferential set-up (likelihood, prior, values of hyperparameters) is the same as in the previous example.

Results We find that for a value of $\beta^* = 1.06$, $d_{1/2}^* \approx \tilde{d}_{1/2}(\beta^*)$. The values of the empirically estimated discrepancies are defined in table 7.

	Mean
Well-specified threshold $d_{1/2}^*$	12.4181
Standard Bayes $\hat{d}_{1/2}$	21.3580
β -Bayes $\tilde{d}_{1/2}(\beta^*)$, $\beta^* = 1.06$	12.4164

Table 7: Mean pairwise posterior discrepancies $d_{1/2}$ across $M = 1000$ independent dataset pairs. $\beta^* = 1.06$ is chosen so that $\tilde{d}_{1/2}(\beta^*) = d_{1/2}^*$.

Downstream performance For this dataset we also investigate the effects on stability of parameter inference, predictions, and predictive accuracy. The metrics used are the same as in section 4.3.

	Standard Bayes	$\beta = 1.1$	$\beta^* = 1.06$	$\beta = 1.3$	$\beta = 1.5$
Mean L2 distance	0.4911	0.3350	0.3786	0.3055	0.3494
Mean Frobenius norm	0.0056	0.0115	0.0058	0.0206	0.0270

Table 8: Mean L2 distance between posterior means and Frobenius norm between posterior covariances across blocks for Standard Bayes and β -Bayes at various β values.

Model	$\bar{d}_{1/2}(\beta)$
Standard Bayes	1.5970
$\beta = 1.1$	1.5117
$\beta^* = 1.06$	1.5231
$\beta = 1.3$	1.5920
$\beta = 1.5$	1.7100

Table 9: Mean $d_{1/2}(\beta)$ between posterior predictives for each model.

Model	MLPD	RMSE	Cov90	Cov95	Cov99	CRPS
Standard Bayes	-1.3306	0.9141	0.936	0.946	0.961	0.4333
(SE)	0.0179	0.0160	0.002	0.002	0.002	0.0051
$\beta = 1.1$	-1.6491	0.9200	0.902	0.917	0.934	0.4028
(SE)	0.0493	0.0165	0.002	0.002	0.002	0.0059
$\beta^* = 1.06$	-1.4234	0.9164	0.918	0.930	0.946	0.4102
(SE)	0.0335	0.0164	0.003	0.003	0.002	0.0056
$\beta = 1.3$	-2.2605	0.9269	0.874	0.897	0.917	0.4028
(SE)	0.0772	0.0164	0.003	0.003	0.002	0.0061
$\beta = 1.5$	-2.1389	0.9270	0.880	0.899	0.919	0.4025
(SE)	0.0715	0.0163	0.003	0.003	0.002	0.0061

Table 10: Predictive performance metrics for each model.

In this example we can see only a slight improvement in predictive accuracy (CRPS), as in the previous case. Both the stability of parameter inference and predictive inference improves under β -Bayes with β^* , as compared to standard Bayes, but not as significantly as for California Housing. A possible reason is that the Gaussian likelihood is misspecified in different ways for the two datasets. Investigating under which types of misspecification β -Bayes can help is one of the directions of the future work.

5 Conclusion

In this work we have established a notion of well-calibrated posteriors, and formalized the criterion which needs to hold for the posteriors to be well-calibrated. We have shown, both theoretically and empirically, that this criterion is often violated under model misspecification, under standard Bayes. We proposed a methodology based on generalized Bayes with β -divergence loss, which allows to make posteriors well-calibrated. This is

only the first step in creating a technique for calibrating posteriors using β -Bayes. In our future steps, we plan to expand the scope of experiments and investigate for which types of model misspecifications the method works. We have already shown in Section 3.5 that β -Bayes cannot improve calibration for some types of misspecification. Therefore, a further direction is to establish the types of misspecification where β -Bayes can be applied. Additionally, in our future work we plan to consider discrepancies other than $d_{1/2}$.

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A Proof of Lemma 1

Proof. By assumption (A1.1), $\tilde{\Pi}_X \xrightarrow{P} \phi_X$ and $\tilde{\Pi}_Y \xrightarrow{P} \phi_Y$ in total variation. Since the Bhattacharyya coefficient $\text{BC}(\cdot, \cdot) = \int \sqrt{\pi_X \pi_Y} d\theta$ is continuous with respect to total variation distance, the continuous mapping theorem gives

$$\text{BC}(\tilde{\Pi}_X, \tilde{\Pi}_Y) \xrightarrow{P} \text{BC}(\phi_X, \phi_Y).$$

By (12), $\text{BC}(\phi_X, \phi_Y) > 0$ for the two Gaussians with the same covariance. Moreover, $g(t) = -2 \log t$ is continuous on $(0, 1]$. Therefore, applying the continuous mapping theorem again gives

$$d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y) = -2 \log \text{BC}(\tilde{\Pi}_X, \tilde{\Pi}_Y) \xrightarrow{P} -2 \log \text{BC}(\phi_X, \phi_Y) = d_{1/2}(\phi_X, \phi_Y).$$

As

$$d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y) \xrightarrow{P} d_{1/2}(\phi_X, \phi_Y)$$

and $d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y)$ is uniformly integrable by assumption (A1.3), Vitali's theorem implies

$$d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y) \xrightarrow{L^1} d_{1/2}(\phi_X, \phi_Y).$$

Therefore, by definition of convergence in L^1 and the standard inequality $|\mathbb{E}[Z]| \leq \mathbb{E}[|Z|]$ we get

$$\left| \mathbb{E}\{d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y)\} - \mathbb{E}\{d_{1/2}(\phi_X, \phi_Y)\} \right| \leq \mathbb{E}\left[\left| d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y) - d_{1/2}(\phi_X, \phi_Y) \right| \right] \rightarrow 0,$$

and

$$\mathbb{E}\{d_{1/2}(\tilde{\Pi}_X, \tilde{\Pi}_Y)\} \rightarrow \mathbb{E}\{d_{1/2}(\phi_X, \phi_Y)\}.$$

□

B Proof of Lemma 2

Proof. Expanding (23) and (24), we identify six terms:

$$E_{P_\circ} [p(x | \theta_*^{(\beta)})^{2(\beta-1)} \nabla_\theta \log p(x | \theta_*^{(\beta)}) \nabla_\theta \log p(x | \theta_*^{(\beta)})^\top], \quad (17)$$

$$E_{P_\circ} [p(x | \theta_*^{(\beta)})^{\beta-1} \nabla_\theta \log p(x | \theta_*^{(\beta)})] E_{P_\circ} [p(x | \theta_*^{(\beta)})^{\beta-1} \nabla_\theta \log p(x | \theta_*^{(\beta)})]^\top, \quad (18)$$

$$E_{P_\circ} [p(x | \theta_*^{(\beta)})^{\beta-1} \nabla_\theta^2 \log p(x | \theta_*^{(\beta)})], \quad (19)$$

$$E_{P_\circ} [p(x | \theta_*^{(\beta)})^{\beta-1} \nabla_\theta \log p(x | \theta_*^{(\beta)}) \nabla_\theta \log p(x | \theta_*^{(\beta)})^\top], \quad (20)$$

$$\int p(z | \theta_*^{(\beta)})^\beta \nabla_\theta^2 \log p(z | \theta_*^{(\beta)}) dz, \quad (21)$$

$$\int p(z | \theta_*^{(\beta)})^\beta \nabla_\theta \log p(z | \theta_*^{(\beta)}) \nabla_\theta \log p(z | \theta_*^{(\beta)})^\top dz. \quad (22)$$

We fix an arbitrary $\beta_0 \in (1, C]$ and let $\beta_n \rightarrow \beta_0$. Continuity of $\text{Tr}(I_{\theta_*^{(\beta)}} J_{\theta_*^{(\beta)}}^{-1})$ at β_0 follows from continuity of each of the six terms (17)–(22), as matrix trace and inversion are continuous operations. Let's note that by assumptions of Theorem 1, the inverse of $J_{\theta_*^{(\beta_0)}}$ exists.

We show the continuity of each term via the Dominated Convergence Theorem (DCT), which requires pointwise convergence and the existence of integrable dominating functions.

Pointwise convergence of integrands (17)–(22). By Remark in 3.3, the assumptions of Theorem 1 hold if the third derivatives of (5) are continuous. This implies the continuity of $\theta \mapsto p(x | \theta)$ on Θ for $P_\circ$ -almost every x , together with its first and second derivatives. By assumption (A2.1), $p(x | \theta_*^{(\beta_0)}) > 0$ for $P_\circ$ -a.e. x , and by assumption (A2.5), $\theta_*^{(\beta_n)} \rightarrow \theta_*^{(\beta_0)}$. Therefore, we get, for $P_\circ$ -a.e. x ,

$$p(x | \theta_*^{(\beta_n)}) \rightarrow p(x | \theta_*^{(\beta_0)}),$$

$$\nabla_\theta \log p(x | \theta) \Big|_{\theta=\theta_*^{(\beta_n)}} \rightarrow \nabla_\theta \log p(x | \theta) \Big|_{\theta=\theta_*^{(\beta_0)}},$$

$$\nabla_\theta^2 \log p(x | \theta) \Big|_{\theta=\theta_*^{(\beta_n)}} \rightarrow \nabla_\theta^2 \log p(x | \theta) \Big|_{\theta=\theta_*^{(\beta_0)}}.$$

Therefore, the positive powers of $p(x \mid \theta_*^{(\beta_n)})$ are also continuous:

$$\begin{aligned} p(x \mid \theta_*^{(\beta_n)})^{\beta_n-1} &\rightarrow p(x \mid \theta_*^{(\beta_0)})^{\beta_0-1}, \\ p(x \mid \theta_*^{(\beta_n)})^{2(\beta_n-1)} &\rightarrow p(x \mid \theta_*^{(\beta_0)})^{2(\beta_0-1)}, \\ p(z \mid \theta_*^{(\beta_n)})^{\beta_n} &\rightarrow p(z \mid \theta_*^{(\beta_0)})^{\beta_0}. \end{aligned}$$

Thus, every integrand in (17)–(22) converges to its value at β_0 .

Dominating functions. For $\beta \in (1, C]$, the following bounds hold:

$$p(x \mid \theta)^{2(\beta-1)} \leq \max(1, p(x \mid \theta)^{2(C-1)}), \quad p(x \mid \theta)^{\beta-1} \leq \max(1, p(x \mid \theta)^{C-1}),$$

$$p(z \mid \theta)^\beta \leq \max(p(z \mid \theta), p(z \mid \theta)^C).$$

Applying these bounds, the dominating terms for (17)–(22) are as follows.

- Term (17) is dominated by $\max(1, p(x \mid \theta)^{2(C-1)}) \|\nabla_\theta \log p(x \mid \theta)\|^2$, which is integrable by (A2.2).
- Terms (19), (20) are dominated by $\max(1, p(x \mid \theta)^{C-1}) (\|\nabla_\theta^2 \log p(x \mid \theta)\| + \|\nabla_\theta \log p(x \mid \theta)\|)^2$, which is integrable by (A2.3).
- Terms (21), (22) are dominated by $\max(p(z \mid \theta), p(z \mid \theta)^C) (\|\nabla_\theta^2 \log p(z \mid \theta)\| + \|\nabla_\theta \log p(z \mid \theta)\|)^2$, which is integrable by (A2.4).

Term (18) requires dominating $\mathbb{E}_{P_\circ} [p(x \mid \theta)^{\beta-1} \|\nabla_\theta \log p(x \mid \theta)\|]$. Applying Cauchy–Schwarz inequality, we get

$$\mathbb{E}_{P_\circ} [\max(1, p(x \mid \theta)^{C-1}) \|\nabla_\theta \log p(x \mid \theta)\|]^2 \leq \mathbb{E}_{P_\circ} [\max(1, p(x \mid \theta)^{C-1})^2 \|\nabla_\theta \log p(x \mid \theta)\|^2],$$

which, using $\max(1, p(x \mid \theta)^{C-1})^2 = \max(1, p(x \mid \theta)^{2(C-1)})$, gives

$$\mathbb{E}_{P_\circ} [p(x \mid \theta)^{\beta-1} \|\nabla_\theta \log p(x \mid \theta)\|]^2 \leq \mathbb{E}_{P_\circ} [\max(1, p(x \mid \theta)^{2(C-1)}) \|\nabla_\theta \log p(x \mid \theta)\|^2] < \infty$$

by (A2.2). Therefore, applying DCT to each of the terms (17)–(22), we get continuity

of $I_{\theta_*^{(\beta)}}$ and $J_{\theta_*^{(\beta)}}$ at β_0 . Since $\beta_0 \in (1, C]$ was chosen arbitrarily, this holds for all $\beta \in (1, C]$. $\square$

C Exact formulas

Formulas for $I_{\theta_*^{(\beta)}}$, $J_{\theta_*^{(\beta)}}$

$$I_{\theta_*^{(\beta)}} = \mathbb{E}_{P_\circ} \left[p(x|\theta_*^{(\beta)})^{2(\beta-1)} \nabla_{\theta_*^{(\beta)}} \log p(x|\theta_*^{(\beta)}) \nabla_{\theta_*^{(\beta)}} \log p(x|\theta_*^{(\beta)})^\top \right] \\ - \mathbb{E}_{P_\circ} \left[p(x|\theta_*^{(\beta)})^{\beta-1} \nabla_{\theta_*^{(\beta)}} \log p(x|\theta_*^{(\beta)}) \right] \mathbb{E}_{P_\circ} \left[p(x|\theta_*^{(\beta)})^{\beta-1} \nabla_{\theta_*^{(\beta)}} \log p(x|\theta_*^{(\beta)}) \right]^\top, \quad (23)$$

and

$$J_{\theta_*^{(\beta)}} = -\mathbb{E}_{P_\circ} \left[p(x|\theta_*^{(\beta)})^{\beta-1} \left(\nabla_{\theta_*^{(\beta)}}^2 \log p(x|\theta_*^{(\beta)}) + (\beta-1) \nabla_{\theta_*^{(\beta)}} \log p(x|\theta_*^{(\beta)}) \nabla_{\theta_*^{(\beta)}} \log p(x|\theta_*^{(\beta)})^\top \right) \right] \\ + \int p(z|\theta_*^{(\beta)})^\beta \left(\nabla_{\theta_*^{(\beta)}}^2 \log p(z|\theta_*^{(\beta)}) + \beta \nabla_{\theta_*^{(\beta)}} \log p(z|\theta_*^{(\beta)}) \nabla_{\theta_*^{(\beta)}} \log p(z|\theta_*^{(\beta)})^\top \right) dz. \quad (24)$$

Parameters of NIG posterior

$$\mathbf{V}_{ji} = (\Sigma_0^{-1} + \mathbf{X}_{ji}^\top \mathbf{X}_{ji})^{-1}, \\ \boldsymbol{\mu}_{ji} = \mathbf{V}_{ji} \mathbf{X}_{ji}^\top \mathbf{y}_{ji}, \\ a_n = a_0 + \frac{n}{2}, \\ b_{ji} = b_0 + \frac{1}{2} (\mathbf{y}_{ji}^\top \mathbf{y}_{ji} - \boldsymbol{\mu}_{ji}^\top \mathbf{V}_{ji}^{-1} \boldsymbol{\mu}_{ji}).$$

Observed discrepancy between NIG posteriors Using the densities of NIG posteriors, we get

$$\log BC(\hat{\pi}_{1i}, \hat{\pi}_{2i}) = \frac{a_n}{2} (\log b_{1i} + \log b_{2i}) - \frac{1}{4} (\log |\mathbf{V}_{1i}| + \log |\mathbf{V}_{2i}|) + \frac{D}{2} \log 2 - \frac{1}{2} \log |S_i| - a_n \log \lambda_i,$$

where

$$\lambda_i = \frac{b_{1i} + b_{2i}}{2} + \frac{1}{4} \left(\boldsymbol{\mu}_{1i}^\top \mathbf{V}_{1i}^{-1} \boldsymbol{\mu}_{1i} + \boldsymbol{\mu}_{2i}^\top \mathbf{V}_{2i}^{-1} \boldsymbol{\mu}_{2i} - \mathbf{h}_i^\top S_i^{-1} \mathbf{h}_i \right),$$

$$S_i = \mathbf{V}_{1i}^{-1} + \mathbf{V}_{2i}^{-1}, \text{ and } \mathbf{h}_i = \mathbf{V}_{1i}^{-1} \boldsymbol{\mu}_{1i} + \mathbf{V}_{2i}^{-1} \boldsymbol{\mu}_{2i}.$$

The observed discrepancy can be computed as

$$\hat{d}_{1/2} = \frac{1}{M} \sum_{i=1}^M -2 \log BC(\hat{\pi}_{1i}, \hat{\pi}_{2i}).$$